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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

SULFURIC ANODIZING

TYPE II

ANODIZE CLASS **TYPE II**

TYPE I	TYPE II	TYPE III
CHROMIC · THIN	SULFURIC · ~5–25 MM	HARDCOAT · THICK

SULFURIC · DECORATIVE & PROTECTIVE · ~5–25 MM

THE COMPLETE TRAINING MANUAL

*A full training course for the brand-new operator — from "wait, the part is the positive electrode?" to a signed certificate of completion. Standard decorative/protective sulfuric anodize on aluminum. We don't deposit metal here — we **grow** it out of the part. Assumes you know nothing. Teaches you everything.*

01 INSP/RACK ▸ 02 CLEAN ▸ 03 ETCH ▸ 04 RINSE ▸ 05 DESMUT ▸ 06 RINSE

07 ANODIZE ▸ 08 RINSE ▸ 09 DYE ▸ 10 SEAL ▸ 11 RINSE/DRY ▸ 12 INSPECT

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

FIND YOUR WAY

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, your customer’s spec (MIL-PRF-8625, AMS, ASTM, etc.), or your supervisor’s instructions. When this manual and your shop’s official procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN EXACTLY THE RIGHT PLACE — AND HERE, "ANODE" IS THE WHOLE GAME

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **sulfuric acid anodizing line**. Maybe you started this week. Maybe you've been racking parts for a year and finally want to know *why* you do the things you do. Either way, you're in the right place, and we're glad you're here.

Here is the first big thing to wrap your head around, and it's going to feel backwards if you've ever seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a tough, glass-hard oxide layer *out of the aluminum itself.*** And to do that, your part is hooked up as the **anode** — the *positive* electrode — which is the exact opposite of plating. If that sentence made your eyebrows go up, good. That's the single biggest idea in this whole book, and we'll teach it from the ground up.

The second thing we'll tell you up front: **this manual assumes you know nothing about anodizing, chemistry, or electricity.** That's not an insult — it's a promise. Every term gets defined the first time we use it; every step explains not just *what* to do but *why* it matters. It's written in the spirit of the *plain-language* guides — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of acid.

> ★ **REMEMBER**
> You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect and rack it, clean it, etch it, rinse it, desmut it, rinse it, **anodize it**, rinse it, (optionally) **dye it**, rinse it, **seal it**, rinse/dry it, and inspect it. Reading it in order matters, because (as you'll soon learn) **the order of an anodizing line is not random** — and neither is the order of this book.

> 💡 **TIP**
> Read the Fundamentals (Part One) all the way through *before* you read any individual station chapter. The station chapters assume you already understand the big picture Part One gives you — especially the "the part is the anode and we grow oxide out of it" idea. It's the difference between reading a recipe when you already know how to cook versus the very first time you've stepped into a kitchen.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

> 💡 **TIP**
> A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a 20-year veteran would lean over and tell you. These save you time and headaches.

> ★ **REMEMBER**
> A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.

> ⚠ **HEADS-UP**
> A safety warning. Read every single one. These keep your skin, eyes, and lungs intact. An anodize line has strong acid, hot caustic, and near-boiling tanks — we use these *liberally*.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line. Read them and you'll understand it on a deeper level.

> ☀ **FUN FACT**
> A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (sometimes paired with a red "Top Mistakes" card) — the things you must be able to *say out loud, unprompted,* before you're cleared on that station — and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

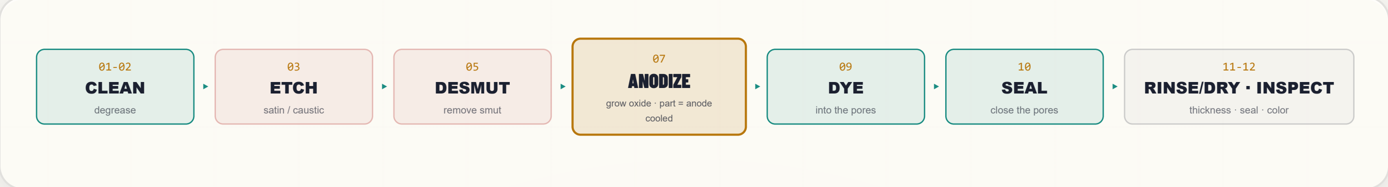
1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **Describe what Type II sulfuric anodizing does** and why a shop chooses it — a hard, integral, corrosion-resistant, electrically insulating oxide that can be **dyed** beautiful colors and **sealed** to lock them in.
3. **Explain the porous oxide structure** — why anodic oxide is full of microscopic pores, why those pores let us dye, and why we then have to **seal** them.
4. **Explain dimensional growth** — the coating grows roughly **half into** the surface and **half out**, so a part gets bigger by about half the coating thickness per surface.
5. **Place Type II in the family** — understand how Type I (chromic), Type II (sulfuric), and Type III (hardcoat) differ in chemistry, thickness, and use.
6. **Explain why aluminum alloy matters** — how copper, silicon, and cast vs. wrought structure change the color, clarity, and quality of the anodize.
7. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
8. **Recognize the main hazards** at each station — sulfuric acid, hot caustic etch, hydrogen gas at the cathode, DC electrical/arcing, near-boiling seal tanks, and dye handling — and know the correct PPE and emergency response.
9. **Use the basic field checks** every anodizer relies on — water-break test, coating thickness (eddy-current), and seal-quality tests (dye-stain, admittance).
10. **Spot trouble early** and **talk the language** with your lead, your chemist, and your supplier.

★ **REMEMBER**

You are not expected to hit all of these on day one. Anodizing is a craft. The goal is steady competence, not instant mastery — but the *safety* objectives (#8) are not optional and not gradual. You must have those before you work the line at all.

• THE LINE AT A GLANCE

The whole Type II line in one strip. Don't memorize yet — feel the rhythm: **Inspect/Rack → Clean → Etch → Rinse → Desmut → Rinse → ANODIZE → Rinse → (Dye) → Rinse → Seal → Rinse/Dry → Inspect.**



★ **REMEMBER**

Notice the pattern: **a rinse follows almost every working tank.** Rinses aren't breaks — they're checkpoints that keep one chemistry from poisoning the next. On an anodize line a bad rinse before the anodize tank or before the seal tank can quietly wreck the whole job. They aren't filler; they're some of the most important steps you'll perform.

☀ **FUN FACT**

The aluminum oxide we grow in this tank is chemically the same family as **sapphire and ruby** (crystalline aluminum oxide, Al_2O_3). You're not bolting on a coating — you're turning the skin of the part into a ceramic that's part of the metal.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a hard, protective oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. Now let's unpack it, because it's genuinely different from plating and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you take your part, hook it up as the **cathode** (the *negative* electrode), and you *deposit a different metal* onto it from a bath. Zinc goes onto steel. Chrome goes onto steel. The part collects metal that came from somewhere else (dissolving anodes or the bath chemistry). The coating is a *separate metal sitting on top of* your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word "anodize" comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate, because it's integral to the metal.

> ★ **REMEMBER — THE SINGLE BIGGEST IDEA IN THIS BOOK**
> In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself.** If you remember nothing else from Chapter 1, remember this. Everything else follows from it.

HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

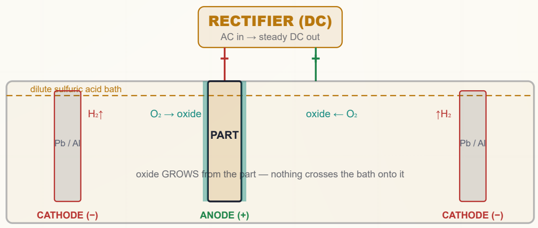
Picture a tank full of liquid called the **electrolyte** (or just "the bath"). For Type II, that liquid is **dilute sulfuric acid** — sulfuric acid mixed with water. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current anodizing needs):

- **The anode** — this is your **aluminum part**, connected to the **positive** terminal. *"Anode = the part, the thing growing oxide."*
- **The cathode** — bars or plates (commonly **lead** or **aluminum 6063**) hanging in the same tank, connected to the **negative** terminal. The cathode just completes the circuit and is where **hydrogen gas** bubbles off.

When you turn on the rectifier, here's the dance that happens:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al_2O_3)** — a hard, ceramic layer.
3. The sulfuric acid is slightly aggressive, so as the oxide forms, the acid is *also* gently dissolving it — and this tug-of-war (oxide forming faster than the acid dissolves it) is exactly what creates the famous **porous structure** (Chapter 3).
4. Over at the cathode, **hydrogen gas** bubbles off harmlessly into the air (this is a safety item — flammable gas — we'll come back to it).

The net result: the surface of your part *converts* into a controlled, porous oxide layer that grows thicker the longer you anodize and the more current you push.



Compare to a plating cell: the labels are reversed — here the **part is the ANODE (+)** and the counter-electrodes are the cathodes.

⚙ TECHNICAL STUFF — THE TWO REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2Al + 3H_2O \rightarrow Al_2O_3 + 6H^+ + 6e^-$. At the cathode, those electrons reduce hydrogen ions: $6H^+ + 6e^- \rightarrow 3H_2\uparrow$. So **oxygen builds the coating at your part; hydrogen gas leaves at the counter-electrode.** No aluminum travels through the bath onto something else — it stays put and turns to oxide in place.

⚙ TECHNICAL STUFF — WHY THE BATH MUST BE *COOLED*

This reaction is **exothermic** (it makes heat), and a lot of electrical energy turns to heat right at the part surface. If the bath gets too warm, the acid dissolves the oxide too aggressively — you get a soft, powdery, "burned," or thin coating instead of a hard, clear one. That's why a Type II tank has **refrigerated cooling and strong air agitation** to hold it near ~20 °C (68 °F). On this line, *cold and well-stirred is quality.*

☀ FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it. That's why anodized aluminum can't "chip off" to bare metal the way paint or a bad plate can — there's no glue line. The oxide and the metal are continuous.

CHAPTER 2

WHAT IS "TYPE II"?

THE EVERYDAY WORKHORSE ANODIZE — PROTECTIVE, DYEABLE, AND ON EVERYTHING FROM COOKWARE TO AIRCRAFT

Type II anodizing is the standard **sulfuric-acid anodize**: a controlled aluminum-oxide layer, typically **~5–25 µm (0.0002–0.001″)** thick, grown in a dilute sulfuric acid bath at room temperature. It's the most common anodize in the world. The "Type II" name comes from the U.S. spec **MIL-A-8625 / MIL-PRF-8625**, which sorts anodize into Type I (chromic), Type II (sulfuric), and Type III (hardcoat).

• WHY SHOPS CHOOSE IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Corrosion resistance	The integral oxide protects aluminum from weathering and salt far better than bare metal — especially once **sealed**.
Hardness & wear resistance	The oxide is ceramic-hard — far tougher than soft aluminum (though not as thick/hard as Type III hardcoat).
It can be DYED	The porous oxide soaks up dye like a sponge, giving rich, durable colors that are *inside* the coating, not painted on.
Electrically insulating	Anodic oxide is a good insulator — useful for electronics, busbars (with masking), and hardware.
Integral & non-flaking	Because it's grown from the metal, it won't peel or chip to bare aluminum.
Good paint/adhesive base	The porous surface (unsealed or lightly sealed) is an excellent primer for paint and bonding.

WHERE YOU'LL SEE IT

Cookware, flashlights, laptop and phone housings, architectural trim, cosmetic packaging, hand tools, firearms accessories, bicycle parts, and a huge amount of aerospace and military hardware (where the spec callout matters a great deal).

★ **REMEMBER**

Type II's two superpowers are **(1) it protects** and **(2) it takes color.** The first comes from the hard integral oxide; the second comes from the *porous* structure that lets dye in. The whole back half of the line — dye and seal — exists to use that second superpower and then lock it in.

💡 **TIP**

When someone says "anodize" with no type, they almost always mean **Type II sulfuric.** It's the default. If a print says "Type I," "hardcoat," "Type III," or a specialty like "BSAA," that's a *different* process — check the spec and ask, because chemistry, thickness, and color all change.

☀ **FUN FACT**

The bright, candy-colored aluminum carabiners, pens, and tent stakes you've seen are almost all **dyed Type II anodize.** The color lives *inside* microscopic pores in the oxide and is then sealed shut — which is why it doesn't rub off like paint.

CHAPTER 3

THE POROUS OXIDE

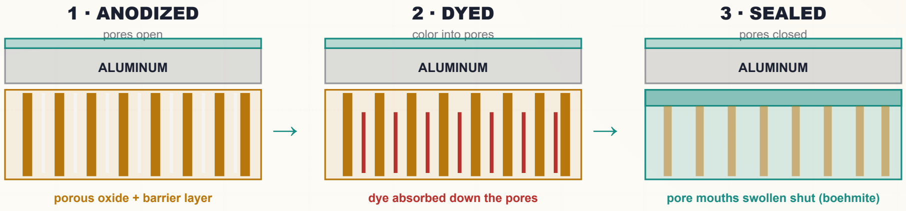
THE STRANGEST, MOST IMPORTANT IDEA ON THE LINE — AND THE REASON ANODIZING CAN BE ANY COLOR YOU WANT

This is the chapter that makes anodizing *anodizing*. Slow down here.

THE OXIDE GROWS WITH MICROSCOPIC HOLES IN IT

Remember the tug-of-war from Chapter 1: oxide forms at the surface while the sulfuric acid gently dissolves it. The result is not a smooth glassy sheet — it's a layer made of **billions of tiny columns with a microscopic pore down the center of each one**, like a honeycomb of straws standing up on the metal. Underneath those pores sits a thin, dense **barrier layer** right against the aluminum.

So the finished anodize has two parts: a thin, dense **barrier layer** at the bottom (against the metal), and a thick **porous layer** above it, full of vertical pores open to the surface. Those open pores are the whole reason anodizing is special — because **anything you can get *into* the pores becomes part of the coating.**



STEP ONE: DYE (COLOR GOES INTO THE PORES)

Right after anodizing, the pores are open and thirsty. Dip the part in a **dye bath** and the dye is drawn down into those pores, coloring the oxide from the inside. The color isn't a layer on top — it's *absorbed into* the structure of the coating. That's why dyed anodize is so durable compared to paint.

STEP TWO: SEAL (CLOSE THE PORES TO LOCK IT IN)

Here's the catch: open pores are also a weakness. They can hold contamination, they'll let dye bleed or fade, and they leave the part less corrosion-resistant. So the **last** chemical step is **sealing** — we close the pores up. Sealing converts the pore walls into a swollen hydrated oxide (and/or fills them) so the pore mouths pinch shut. The classic way is **hot water near boiling**, which makes the oxide swell and "heal" the pores closed.

★ **REMEMBER — THE HEART OF ANODIZING**

Anodize → (Dye) → Seal. Anodize grows the porous oxide. Dye puts color *into* the pores. Seal *closes* the pores to lock in the color and corrosion resistance. This "dye-then-seal" idea exists in *no other* finishing process you've learned — it's unique to anodizing, and it's the soul of this line.

⚠ **HEADS-UP — SEALING IS A ONE-WAY DOOR**

Once you seal, you **cannot dye** anymore — the pores are shut. So the order is non-negotiable: dye *before* seal, never after. Seal a part you forgot to dye and the only fix is to strip and re-anodize. (And a half-sealed part dyes blotchy or not at all.)

⚙ **TECHNICAL STUFF — WHY HOT-WATER SEALING WORKS**

Near-boiling water converts the aluminum oxide at the pore walls into **boehmite** (a hydrated aluminum oxide, AlOOH) that takes up more volume than the original oxide. The pore walls swell and the pore mouths close. That volume increase is literally what "seals" the coating. Mid-temp nickel-acetate and cold nickel-fluoride seals add metal salts that precipitate in the pore mouths to help plug them, which is why they can work faster or cooler.

☀ **FUN FACT**

An unsealed, freshly anodized part is so absorbent that you can write on it with a permanent marker and the "ink" soaks in like the dye does. Veterans sometimes use a dye-spot or marker test to check that the pores are still open before sealing.

CHAPTER 4

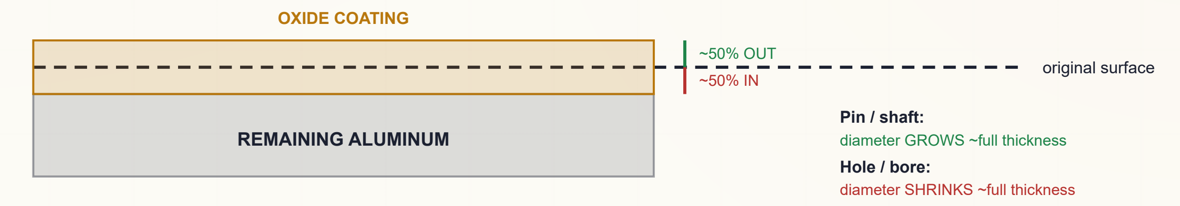
DIMENSIONAL GROWTH

THE COATING MAKES THE PART BIGGER — AND YOU HAVE TO PLAN FOR IT

Because we're converting aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the coating grows in **two directions at once**:

- About **half of the coating thickness grows *into*** the original surface (consuming aluminum).
- About **half grows *out*** from the original surface (the part gets bigger).

So for a coating that's, say, **20 µm (0.0008″) thick**, the surface moves outward by roughly **10 µm (0.0004″) per face**. On a shaft or a pin, that means the **diameter grows by about the full coating thickness** (because there are two opposing faces), and a hole's **diameter shrinks by about the full coating thickness.**



★ REMEMBER — THE 50% RULE

Anodize grows **roughly half in, half out.** Each *surface* moves outward by about **half** the coating thickness. On a *diameter* (two faces), the total change is about the **full** coating thickness — pins grow, holes shrink. Tight-tolerance parts must account for this *before* anodizing.

💡 TIP

When a machinist asks "how much do I undersize for anodize?", the rule of thumb is: subtract about **half the spec'd coating thickness per surface** from the pre-anodize dimension on critical features. For threads and press-fits, this matters a lot. Always confirm with the print and your shop's standard.

⚠ HEADS-UP — MASKING AND THREADS

Areas that must *not* grow (threaded holes, bearing fits, electrical-contact points, grounding surfaces) are usually **masked off** so no oxide forms there — or are machined oversize/undersize on purpose. Anodize is also **electrically insulating**, so any surface that needs to conduct (a ground path) must be masked or it won't work after anodizing.

⚙ TECHNICAL STUFF — THE PILLING–BEDWORTH IDEA

Aluminum oxide occupies more volume than the aluminum it formed from (the "Pilling–Bedworth ratio" for Al_2O_3 is greater than 1). That volume expansion is *why* the coating grows outward at all, and it's the same expansion that lets hot-water sealing pinch the pores shut. One simple fact — oxide is bulkier than metal — explains both dimensional growth *and* sealing.

☀ FUN FACT

On Type III hardcoat (a much thicker cousin of Type II), dimensional growth is big enough that aerospace drawings often call out the *pre-anodize* and *post-anodize* dimensions separately. Type II is thinner, but the same physics applies — just smaller numbers.

CHAPTER 5

THE TYPE FAMILY

SAME IDEA, THREE FLAVORS — KNOW WHICH ONE YOUR PRINT IS ASKING FOR

All three “types” grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, thickness, and purpose.**

	TYPE I — CHROMIC	TYPE II — SULFURIC *(THIS MANUAL)*	TYPE III — HARDCOAT
Electrolyte	Chromic acid	**Sulfuric acid (~165–200 g/L)**	Sulfuric acid (often more concentrated/colder)
Typical thickness	~0.5–7.5 µm	**~5–25 µm**	~25–100+ µm
Temperature	Warm-ish	**~20–21 °C (68–72 °F)**	Cold (~0–10 °C)
Voltage / current	Lower	**~12–18 V, ~12–18 ASF**	Higher V, higher CD
Color / dyeing	Gray, limited	**Excellent dyeing**	Darkens naturally; can be dyed (bronze/black)
Main use	Aerospace — minimal dim. change & fatigue	**Decorative + protective, everyday workhorse**	Maximum hardness & wear (cylinders, gears, sliders)

★ **REMEMBER**

Type I = chromic, thin, aerospace/fatigue. Type II = sulfuric, medium, decorative + protective (the standard). Type III = sulfuric hardcoat, thick, maximum wear. This manual is Type II — and it’s the base the rest of the family builds on.

💡 **TIP**

Beyond the three “types” there are **specialty anodizes** you’ll meet later in this training series: **BSAA** (boric-sulfuric, a chromic replacement for aerospace), **PAA** (phosphoric-acid, for adhesive bonding), **bright anodize** (chemically/electro-polished first for a mirror look), and **two color methods** — *integral color* (color from the alloy/process itself) and *two-step electrolytic* (metal salts deposited in the pores for very durable bronzes/blacks, common in architecture). Each gets its own manual; they all stand on the Type II fundamentals you’re learning now.

⚙ **TECHNICAL STUFF — WHY TYPE I USES CHROMIC FOR AEROSPACE**

Chromic anodize is thin and its acid is *gentler* on the base metal, so it removes very little aluminum and is kinder to **fatigue life** on stressed aircraft parts. It also leaves any trapped residue less corrosive than trapped sulfuric. The tradeoff is hexavalent-chromium chemistry (a carcinogen), which is exactly why **BSAA** was developed as a sulfuric-based replacement.

CHAPTER 6

ALUMINUM ALLOYS

ANODIZING ONLY WORKS ON ALUMINUM — AND NOT ALL ALUMINUM ANODIZES THE SAME

Anodizing only works on **aluminum** and a handful of other “valve metals” (titanium, magnesium, tantalum, niobium — each with its own process). On this line, it’s aluminum. But “aluminum” isn’t one thing — it’s dozens of **alloys**, and the alloying elements dramatically change how the part anodizes.

WROUGHT VS. CAST

- **Wrought alloys** (rolled/extruded sheet, plate, bar, extrusion — the 1xxx–7xxx series) generally anodize **cleanly and predictably.**
- **Cast alloys** (die-cast, sand-cast) usually contain **lots of silicon** and other elements and often anodize **gray, mottled, or dull** — and can be hard to color evenly. Customers are often surprised that a cast part won’t match a wrought part.

THE BIG TROUBLEMAKERS: SILICON AND COPPER

- **Silicon (Si):** Silicon particles don’t convert to clear oxide — they leave the coating **gray to dark/charcoal** and cloudy. High-silicon casting alloys are the classic “why is my part gray?” culprit.
- **Copper (Cu):** Copper-rich alloys (the **2xxx series**, e.g., 2024 — common in aerospace) anodize less brightly, can give a **yellowish or dull** coating, and are more prone to defects. They still anodize, but appearance and corrosion performance suffer.

THE FRIENDLY ALLOYS

5xxx (magnesium) and **6xxx (magnesium-silicon, e.g., 6061/6063)** anodize **very well** and are the everyday favorites for clear and dyed work. **6063** (“architectural”) anodizes especially clean and bright. **7xxx (zinc)** anodizes reasonably but can vary.

ALLOY FAMILY	MAIN ALLOYING ELEMENT	ANODIZE BEHAVIOR
1xxx (pure Al)	none (99%+ Al)	Excellent, very clear/bright
2xxx	Copper	Dull/yellowish, harder to anodize well
5xxx	Magnesium	Very good, clear
6xxx (6061/6063)	Mg + Si	Excellent — the workhorse
7xxx	Zinc	Good, can vary
Cast (high-Si)	Silicon	Gray/mottled, hard to color

★ **REMEMBER**

The alloy decides the look. The cleanest, brightest, best-coloring anodize comes from low-alloy and 5xxx/6xxx wrought aluminum. **Copper (2xxx) and silicon (cast) are the two elements that dull and gray the coating.** If a customer wants a bright dyed finish, the alloy matters as much as the line.

💡 **TIP**

Always know your alloy *before* you promise a color. A brilliant red on 6063 may come out muddy on a high-silicon casting or a 2024 part. When in doubt, anodize a sample of the *actual* alloy and lot first.

☀ **FUN FACT**

Pure aluminum and high-purity 6063 anodize so clearly that the architectural industry built whole product lines (window frames, storefronts) around the consistent, beautiful clear and bronze anodize they produce.

CHAPTER 7

HOW THE LINE WORKS

WALKING THE LINE — EVERY TANK HAS A JOB, AND THE ORDER IS THE RECIPE

A Type II line is a sequence of tanks. Here's the whole journey and *why each step has to be where it is.*

1. **Inspect / Rack** — Check the part and clamp it tight to a rack. Anodizing is all about **electrical contact** (the part *is* the anode), so a loose part = no coating or a burned part. The contact point won't anodize, so it's chosen carefully.
2. **Clean / Degrease** — Remove oils and shop soil. Oxide won't grow evenly through grease.
3. **Etch (caustic)** — A hot caustic dip that removes the natural oxide skin, evens the surface, and gives the matte **satin finish** most anodize has. It also removes a little aluminum.
4. **Rinse** — Wash off caustic before it contaminates the acid steps.
5. **Desmut / Deox** — Etching leaves a dark **smut** (the alloying elements — copper, silicon, iron — that didn't dissolve). An acid desmut removes it so the surface is bright and uniform before anodizing.
6. **Rinse** — Clean before the main tank.
7. **ANODIZE** — The main event. Part = anode, dilute sulfuric acid, cooled, DC on. Grow the porous oxide.
8. **Rinse** — Gently rinse acid out of the fresh, open pores (don't contaminate the dye/seal).
9. **Dye (optional)** — Color into the open pores.
10. **Rinse** — Remove surface dye so it doesn't bleed.
11. **Seal** — Close the pores to lock in color and corrosion resistance.
12. **Rinse / Dry** — Final clean-up and gentle dry.
13. **Inspect** — Thickness, seal quality, color.

★ **REMEMBER — WHY THE ORDER CAN'T MOVE**

You can't dye before you anodize (no pores yet). You can't seal before you dye (sealed pores won't take color). You can't anodize over grease or smut (uneven, defective oxide). The line *is* the recipe, and like a recipe, the steps go in one order for a reason.

⚠ **HEADS-UP — THE LINE IS A TOUR OF HAZARDS**

As you walk the line you pass **hot caustic** (etch), **strong acids** (desmut, anodize), **flammable hydrogen gas** (cathode), **high-current DC** (rectifier/contacts), **near-boiling water** (seal), and **dyes**. Each station chapter calls out its own. Respect each tank for what *it* is.

💡 **TIP**

When you're learning, walk the empty line with your trainer and just name each tank's *job* and *main hazard* out loud. If you can do that for all twelve, the rest of the manual is mostly detail.

CHAPTER 8

PPE — YOUR DAILY ARMOR

WHAT YOU WEAR, AND WHY EACH PIECE IS THERE

An anodize line works with strong sulfuric acid, hot caustic etch, near-boiling seal tanks, high-current DC, chemical mist, and dyes. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

HAZARD ON THE LINE	PPE THAT PROTECTS YOU
Sulfuric acid (anodize, desmut) — burns, splash	Sealed chemical **splash goggles** + **face shield**, chemical-resistant **gloves**, **apron/suit**, chemical-resistant **boots**
Hot caustic etch — burns worse than you'd expect	Same as above; caustic burns are deep and slow to feel at first
Near-boiling seal tank (~96–100 °C) — scald/steam	Face shield, heat-resistant gloves, long sleeves; mind the steam
Acid mist / fumes	Local exhaust at tanks; respiratory protection per your program
DC electrical / arcing at contacts	Dry hands, dry gloves, no jewelry, follow LOTO
Dyes (staining, some sensitizers)	Gloves; avoid skin/eye contact; check the dye SDS
Slips on wet floors	Non-slip chemical-resistant boots

TIP

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

REMEMBER

On an anodize line, your **eyes and skin** are the most exposed. Strong acid *and* hot caustic *and* near-boiling water all live here. **Sealed splash goggles and a face shield are not optional** at the working tanks.

HEADS-UP

Goggles for chemicals, not just safety glasses. Safety glasses stop flying chips; they do **not** stop an acid or caustic splash from getting into your eyes from the side. At every chemical tank, sealed splash goggles (and a face shield when charging, dumping, or handling parts) are required.

HEADS-UP

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting chemicals — especially dyes and nickel-bearing seal salts. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 9

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read.

SITUATION	WHAT TO DO — IMMEDIATELY
Acid or caustic on skin	Flush immediately with water for **at least 15 minutes.** Remove contaminated clothing while flushing. Get medical attention — caustic burns especially can be deep even when they don't hurt at first.
Acid or caustic in eyes	Go straight to the **eyewash**, hold eyes open, flush **at least 15 minutes**, and get medical help. Don't stop early.
Large splash / drench	Use the **safety shower** — full 15 minutes, clothing off.
Hot seal-tank scald	Cool the burn with water; do not pop blisters; get medical attention for anything beyond minor.
Acid spill	Don't play hero — alert others, contain only if trained, use the spill kit, follow your shop's spill plan. Never mix acids and caustics during cleanup.
Electrical / arc at rectifier or contacts	Don't touch — de-energize via the rectifier controls / LOTO. Never reach into a tank with the current on.
Hydrogen gas	Keep ignition sources (flames, sparks, grinding) away from the cathode/tank area; ensure ventilation is running.
Fluoride exposure (fluoride desmut / cold seal)	Fluoride burns are deep, delayed, and systemically toxic — flush, then apply the **specific first-aid** in the SDS (often **calcium gluconate gel**) and get medical help immediately. Ordinary flushing alone may not be enough.

⚠ **HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP**

Always add ACID to WATER, never water to acid. ("Do as you *oughta*: add acid to water.") Adding water to concentrated sulfuric acid can flash-boil and erupt the acid out of the container. Bath make-up/additions are usually authorized-personnel work — but every operator must know this rule.

⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

LOTO is mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it so no one can switch it back on while someone is working on it. "I'll just be a second" is how people get electrocuted.

★ **REMEMBER**

Know where the **nearest eyewash, safety shower, and spill kit** are at *every* station before you start your shift. In an acid/caustic emergency you won't have time to look. **Eyewash and shower stations must always be unblocked.**

CHAPTER 10

THE SAFETY PHILOSOPHY OF AN ANODIZE LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER

1. TWO CORROSIVES, MIRROR-IMAGE.

The **etch is caustic** (high pH); the **desmut and anodize are acid** (low pH). Both burn. Treat high pH and low pH with equal respect — and never let them mix outside a controlled neutralization.

2. THE PART IS LIVE.

Because the part is the anode, there is **DC current at the work** and at the contacts. Wet hands + current + acid is a bad combination. Respect the electrical side as much as the chemical side.

3. HOT IS A HAZARD TOO.

The seal tank runs **near boiling.** Steam scalds, and a near-boiling splash is as dangerous as a chemical one.

4. HYDROGEN IS FLAMMABLE.

The cathode bubbles off hydrogen gas. Ventilation on, ignition sources away.

5. ENGINEERING CONTROLS FIRST, PPE SECOND.

Tank exhaust, cooling, lids, and splash guards do the heavy lifting; PPE is your last layer, not your first.

6. WHEN IN DOUBT, STOP AND ASK.

The unsafe operator isn't the one who asks questions — it's the one who guesses.

THE HAZARD FAMILIES TO INTERNALIZE

1. **Corrosives (acid AND caustic).** Sulfuric (anodize/desmut) and caustic (etch) burn skin and eyes. *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
2. **Hot / thermal.** Near-boiling seal, hot etch, hot cleaner. *Defense:* face shield, heat gloves, steam awareness.
3. **Electrical & gas.** High-current DC; flammable hydrogen at the cathode. *Defense:* LOTO, dry hands, no ignition sources, ventilation.

★ **REMEMBER**

Corrosives first, thermal second, electrical/gas third. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.

☀ **FUN FACT**

Many veteran anodizers can spot a bath problem — a hint of burning, a dye that's gone muddy — before any instrument confirms it. But notice the lesson hidden there: *they got to be veterans by never getting hurt.* Skill and safety grow together.

★ **REMEMBER**

You now have the whole foundation: what anodizing is (and how it's *not* plating), the part-is-the-anode/grow-oxide idea, the porous oxide, dye-then-seal, dimensional growth, the Type family, alloys, how the line works, what to wear, and the safety mindset. Every station chapter builds on this. Turn the page knowing the *why*, and the *how* will make a lot more sense.

PART TWO — STATION BY STATION
STATION 01 OF 12

INSPECTION & RACKING

"IN ANODIZING, THE PART IS THE WIRE. A LOOSE PART IS A DEAD PART."

PURPOSE & THE "WHY"

Inspect the incoming aluminum part, confirm the **alloy and the spec** (Type II, thickness, color, sealed/unsealed), and **rack it** so it makes firm, current-carrying contact and hangs correctly. Because the part **is the anode**, every bit of current that grows your coating flows *through the rack contact into the part.* A loose or corroded contact means high resistance — **no coating, a thin coating, or a burned contact spot** — and possibly a dropped part in an acid tank. Racking also decides *where* the un-anodized contact mark lands, how gas escapes, and how dye/seal solution drains.

★ **REMEMBER — THE CONTACT MARK IS UNAVOIDABLE**

Wherever the rack grips the part, **no oxide grows** (that spot has to carry current, so it can't be insulated by its own coating). So you place the contact where the mark won't matter — a hidden face, an edge, a hole — and you make it **tight.** Choosing the contact point is a real skill, not an afterthought.

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / print	Confirm **alloy**, Type II, thickness, color, sealed/unsealed
Rack material	Titanium or aluminum	Ti racks don't anodize away; Al racks anodize and need stripping
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, no coat, dropped parts
Contact location	Hidden/non-critical area	The contact point won't anodize (bare mark)
Masking	Plugs/tape/lacquer on no-anodize areas	Threads, grounds, bearing fits, press-fits
Orientation	Allow gas escape & drainage	Avoid trapped air pockets (un-anodized voids)

⚠ **HEADS-UP**

Racks and parts have **sharp edges and pinch points**, and you'll be handling them near chemical tanks. Watch your hands. A part that drops because of a weak rack becomes a **splash hazard** in an acid or caustic tank, not just a scrap part.

💡 **TIP — TITANIUM RACKS ARE WORTH IT**

Titanium racks don't anodize (they passivate), so they keep good contact load after load and don't need stripping. Aluminum racks anodize right along with the part — which insulates the contact over time — so they must be **stripped/re-pointed** regularly to keep contact good.

STEP-BY-STEP

1. Read the traveler (alloy, Type II, thickness, color, masked areas). 2. Inspect for scratches/gouges/prior anodize — anodize *reveals* flaws, doesn't hide them. 3. Plan a hidden contact point that can carry full current. 4. Mask no-anodize areas per print. 5. Rack firmly — good metal-to-metal contact, correct orientation. 6. Verify no parts touch each other (shadowing). 7. Handle by the rack from here on.

SIGN-OFF — STATION 01

Teach-Back: Why contact matters more in anodizing than plating (part = anode); why the contact point won't anodize; why titanium racks are preferred; why orientation (gas/drainage) matters; two no-anodize areas that get masked and why.

"I verified the alloy and spec, chose and made a firm contact point in a non-critical area, masked all no-anodize features, racked for gas escape and drainage with no parts touching, and handled parts by the rack."

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 02 OF 12

CLEAN / DEGREASE

"ANODIZE OVER OIL AND YOU'VE GROWN A DEFECT, NOT A COATING."

PURPOSE & THE "WHY"

Get the part **chemically clean** — no oil, grease, fingerprints, or shop soil — so the oxide grows evenly. Oxide can only grow uniformly from clean aluminum; oil and soil cause **uneven anodizing, streaks, and patchy color.** Cleaning is usually a **mild alkaline (non-etching) soak cleaner** — formulated *not* to attack the aluminum (we save the controlled metal removal for the etch). The free, foolproof check is the **water-break test.**

TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part.

```
FAIL - water beads up          PASS - water sheets evenly
+--------------+               +--------------+
|  o   o   o   |  X            |##############|  OK
|    o    o    |               |##############|
+--------------+               +--------------+
Oil remains -> RE-CLEAN        Surface clean -> proceed
```

PARAMETER	RANGE	WHAT TO WATCH
Type	Mild **non-etching alkaline** cleaner	Must not attack aluminum
Temperature	~50–70 °C (120–160 °F)	Per cleaner TDS
Time	~3–10 min	Heavy oil longer
Concentration	Per TDS	Titrate regularly
pH	Mildly alkaline	Still a skin irritant/burn risk
Water-break test	Pass = continuous sheet	The only reliable field check

⚠ HEADS-UP — HOT ALKALINE SOLUTION

Even a "mild" cleaner is hot and alkaline: splashes irritate/burn skin and eyes, and steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.

STEP-BY-STEP

1. Check tank temperature and concentration are in range. 2. Immerse the racked load fully; start the timer; confirm agitation. 3. Withdraw slowly, draining back into the tank. 4. Run the **water-break test** — continuous sheet → proceed; beading → re-clean. 5. Move promptly to the etch (don't let a clean part sit).

SIGN-OFF — STATION 02

Teach-Back: What the cleaner removes (oil/soil) vs. what it must *not* do (attack aluminum — that's the etch); correctly perform/read a water-break test; the caustic/irritant hazard and PPE; why a non-etching cleaner is used here.

"The part passed the water-break test, the cleaner was in range and at temperature, and I wore required PPE."

Trainee: ________________________ Trainer: ________________________ Date: ________________

STATION 03 OF 12

ETCH (CAUSTIC, SATIN)

"WHERE THE PART GETS ITS MATTE 'ANODIZED LOOK' — AND WHERE YOU'LL RESPECT CAUSTIC FOREVER."

PURPOSE & THE "WHY"

Remove the natural oxide skin and a thin layer of aluminum in a **hot caustic (sodium hydroxide) etch**, giving a uniform, matte **satin finish** and a fresh surface for anodizing. Aluminum always carries a thin, uneven natural oxide and fine surface defects. The caustic etch **dissolves a controlled amount of aluminum**, removing that skin, evening out machining marks, and producing the soft, frosty satin appearance most anodized parts have. Etch too little → shiny, blotchy; etch too much → lost dimension and detail.

★ **REMEMBER — ETCH SETS THE TEXTURE**

The **etch decides whether the part looks satin/matte or shinier.** Light etch = more luster (and more visible defects); heavy etch = more matte (and more metal lost). The etch is also where most of your **dimensional loss** before anodizing happens.

PARAMETER	RANGE	WHAT IT DOES / WATCH
Chemistry	**Sodium hydroxide**, often with additives	Dissolves oxide + aluminum
Concentration	~40–60 g/L NaOH	Higher = faster/more aggressive
Temperature	~50–65 °C (120–150 °F)	Hot — burns and speeds etch
Time	~1–5 min	Longer = more matte + more metal loss
Result	Uniform satin + **smut**	Smut is normal — removed at desmut

⚠ **HEADS-UP — HOT CAUSTIC IS ONE OF THE WORST BURNS ON THE LINE**

Caustic burns are **deep, get worse with time, and may not hurt much at first** — which makes people under-react. Full PPE: sealed goggles, face shield, elbow-length chemical gloves, apron, boots. Skin → flush **15+ minutes**; eyes → eyewash 15 min and get medical help immediately. Never assume "it's just a little."

⚠ **HEADS-UP — CAUSTIC + ALUMINUM MAKES HYDROGEN**

Caustic attacking aluminum releases **hydrogen gas** (flammable). Keep ignition sources away and ventilation running.

STEP-BY-STEP

1. Confirm concentration and temperature in range. 2. Immerse; start the timer for the **specified etch time** (controls finish and metal loss). 3. Agitate gently / per procedure. 4. Withdraw and drain back into the tank — the part will look **dark/smutty** (normal). 5. Move **directly to rinse** — don't let caustic dry on the part.

SIGN-OFF — STATION 03

Teach-Back: What the etch removes (oxide + controlled aluminum) and the finish it creates (satin); the link between etch time and appearance / dimensional loss; why the part comes out dark/smutty and what fixes it (desmut); the caustic-burn hazard and first response (15-min flush, get help).

"I etched for the specified time at the correct concentration/temperature, understood the appearance and dimensional effects, moved straight to rinse, and wore full PPE for hot caustic."

Trainee: ____________ Trainer: ____________ Date: ________

STATION 04 OF 12

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

PURPOSE & THE “WHY”

Wash the **caustic film** off the part before it reaches the acidic desmut and anodize chemistry. The etch is caustic (high pH); desmut and anodize are acidic (low pH). Carry caustic drag-out forward and it **neutralizes and contaminates the acid baths**, wastes chemistry, and causes staining. The rinse knocks drag-out down by dilution. Cleanliness is tracked by **conductivity** (µS/cm) — *lower = cleaner.*

💡 **TIP — COUNTER-FLOW RINSING**

The professional approach is a **counter-current** cascade: parts move forward, water flows backward, so the cleanest water touches the part last. It saves enormous water — important on a line with this many rinses.

PARAMETER	RANGE	NOTES
Temperature	`Ambient`	No heating needed
Time	`30–60 s with agitation`	Longer for cavities/blind holes
Water source	`Municipal or DI`	DI preferred near anodize/seal
Conductivity	`< 500 µS/cm (target < 300)`	Monitor at shift start
Flow	`Continuous overflow / counter-current`	Prevents stagnation

⚠ **HEADS-UP**

This rinse carries dragged-in caustic and starts mildly alkaline. Goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

DO

- Check conductivity at shift start; tell your lead if near/above limit.
- Transfer promptly from the etch — don’t let caustic dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting parts drip-dry between etch and rinse.
- Not flushing cavities; skipping the drain step.

SIGN-OFF — STATION 04

Teach-Back: Why carrying caustic into the acid stages causes problems; define drag-out / drag-in; what conductivity tells you and which way is “clean”; the conductivity limit and target.

“Rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”

Trainee: ____________ Trainer: ____________ Conductivity: ________ µS/cm

STATION 05 OF 12

DESMUT / DEOX

"ETCHING UNCOVERS THE ALLOY'S LEFTOVERS. DESMUT WASHES THEM AWAY."

PURPOSE & THE "WHY"

Remove the dark **smut** the etch left behind — the alloying elements (copper, silicon, iron, manganese) that didn't dissolve in caustic — using an **acid desmut/deox**, leaving a bright, uniform surface for anodizing. When caustic dissolves aluminum, the **other elements in the alloy stay behind as a dark film (smut)** on the surface. Anodize over smut and you get **dark streaks, dull patches, and poor color/adhesion.** The desmut chemistry depends on the alloy: simple alloys may need only **nitric** or **sulfuric**; high-silicon/copper alloys often need a **fluoride-bearing (or proprietary) deox** to cut silicon smut.

> ★ **REMEMBER**
> **Smut after etch is normal.** Desmut's whole job is to remove it. A part that goes into the anodize tank still gray/smutty will come out streaky and dull — desmut is the bright-surface gatekeeper.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Nitric and/or sulfuric; fluoride-bearing/proprietary deox for high-Si/Cu	Match to the alloy
Concentration	`Per product TDS`	Titrate regularly
Temperature	`Often ambient-warm`	Per TDS
Time	`~0.5–3 min`	Until smut is gone, no longer
Result	Bright, uniform, active surface	No gray film remaining

> ⚠ **HEADS-UP — STRONG ACID (AND POSSIBLY FLUORIDE)**
> Desmut is acidic; **fluoride-bearing desmuts are especially hazardous** (hydrofluoric-type chemistry can cause deep, delayed burns and is systemically toxic). Know exactly which chemistry your tank uses, read its SDS, wear full acid PPE, and know the **specific first-aid** (fluoride burns may require **calcium gluconate** per your SDS/medical plan). **Add acid to water** for any make-up.

STEP-BY-STEP

1. Confirm which desmut chemistry the tank is (and its specific hazards/first-aid). 2. Check concentration and temperature in range. 3. Immerse for the specified short time — just until smut clears. 4. Withdraw, drain, move to rinse. 5. Don't over-dip — excessive desmut can etch/pit some alloys.

SIGN-OFF — STATION 05

Teach-Back: What smut is and where it came from (alloying elements left by the etch); why anodizing over smut causes defects; that desmut chemistry depends on the alloy (fluoride for silicon); the special hazard/first-aid of a fluoride-bearing desmut.

"I removed the smut with the correct desmut for the alloy, did not over-dip, and understand the specific hazard and first-aid for this tank (including fluoride if applicable)."

Trainee: ____________ Trainer: ____________ Chemistry: ________

STATION 06 OF 12

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE MAIN EVENT.”

PURPOSE & THE “WHY”

Remove desmut acid before the part enters the anodize tank. Carrying desmut drag-out into the anodize tank introduces unwanted ions (and, with a fluoride desmut, **fluoride contamination** that can roughen/pit the anodize). A clean rinse protects the anodize bath chemistry.

PARAMETER	RANGE	NOTES
Temperature	`Ambient`	—
Time	`30–60 s, agitation`	Longer for cavities
Water	`DI preferred`	Cleaner handoff to anodize
Conductivity	`Per shop spec`	Lower = cleaner
Flow	`Counter-current overflow`	—

⚠ HEADS-UP

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the prior tank was a **fluoride** desmut, this rinse water carries fluoride — handle and dispose accordingly.

💡 TIP

Move from this rinse into the anodize tank **promptly and consistently.** A long, variable dwell in air or rinse before anodizing can cause part-to-part color/thickness differences. Consistency is quality.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: Why a clean handoff to the anodize tank matters; what fluoride drag-in could do.

“The part was fully rinsed, conductivity was acceptable, and I moved promptly and consistently to the anodize tank.”

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 07 OF 12 · THE MAIN EVENT

ANODIZE — THE MAIN TANK

WHERE THE ALUMINUM'S OWN SURFACE BECOMES A HARD, POROUS CERAMIC — AND WHERE THE PART IS THE ANODE

Take a breath. Up to now, every station has had one job: get the part clean, even, and ready. **Station 07 is where the anodize is actually created** — the tank that defines this whole line.

The big idea in one sentence: **we run the aluminum part as the *anode* in cold, dilute sulfuric acid and use DC current to grow a hard, porous aluminum-oxide layer out of the part's own surface.** The cast:

- **Anode** — *your aluminum part* (**positive**). Where the oxide grows.
- **Cathode** — **lead or aluminum (6063) plates** (negative). They carry current and bubble off hydrogen; they do **not** feed metal into the part.
- **Electrolyte** — **dilute sulfuric acid (~165–200 g/L, ~15–20% w/v)**, held **cold (~20–21 °C / 68–72 °F)**, with **air agitation** and **refrigerated cooling.**

★ **REMEMBER — THE ANODIZE TANK IS THE MIRROR IMAGE OF A PLATING TANK**

In plating, the part is the **cathode** and metal lands on it. Here the part is the **anode** and **oxide grows out of it.** The counter-electrodes (cathodes) don't supply the coating — all the coating comes from the part's own aluminum plus oxygen from the bath. Forget this and the whole tank behaves like a mystery.

⚠ **HEADS-UP — SULFURIC ACID + DC + HYDROGEN + COOLING, ALL AT ONCE**

This tank holds **strong sulfuric acid** (burns, splash, mist), runs **high-current DC** (shock/arc at contacts), bubbles **flammable hydrogen** off the cathodes, and **self-heats** (cooling must work or the coating burns and the bath overheats). Engineering controls — **exhaust, cooling, agitation, splash control** — must be working before you run it. Full acid PPE. No ignition sources.

WHY TYPE II SULFURIC (AND WHAT MAKES IT PARTICULAR)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Hard, integral oxide	Won't peel; protects and wears well
Porous → dyeable	The pores take color (Station 09)
Corrosion resistance (sealed)	Long outdoor/marine life once sealed
Electrically insulating	Useful for hardware/electronics
Predictable on 6xxx/5xxx	The everyday workhorse finish

…but the same chemistry demands care: it must be **kept cold** (warm bath = soft/powdery/"burned" coating), it needs **steady current** (thickness depends on it), it's sensitive to **chloride contamination** (pitting), and it grows the coating **half into the part** (dimensional growth).

• THE BATH MAKE-UP — KNOW THE RANGES

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Sulfuric acid (H_2SO_4)	~165–200 g/L (~15–20% w/v)	The electrolyte; grows + gently dissolves the oxide
Temperature	~20–21 °C (68–72 °F)	*The* critical control — too warm = soft/powdery/burned
Voltage (DC)	~12–18 V	Driven to hold the target current density
Current density	~12–18 ASF (~1.2–2.0 A/dm²)	Sets growth rate & structure (the “720 rule”)
Coating thickness	~5–25 µm (0.0002–0.001”)	Per spec; controlled by current × time
Time	~30–60 min (typical)	Longer = thicker, within limits
Cathodes	Lead or aluminum 6063	Carry current; bubble H_2; do not feed metal
Agitation	Air agitation	Even temperature & fresh acid at the surface
Cooling	Refrigeration / heat exchanger	Bath self-heats — *must* be cooled
Dissolved aluminum (Al^{3+})	up to ~12–15 g/L (per spec)	Some helps; too much dulls/streaks & slows the bath
Chloride contamination	Very low (≤ ~50 ppm typ.)	Chloride causes **pitting/burning** — watch it

★ **REMEMBER — THE NUMBERS TO CARRY**

Sulfuric ~165–200 g/L, **temp ~20–21 °C (68–72 °F)**, **~12–18 V**, **~12–18 ASF**, **~5–25 µm**, **~30–60 min**, **lead/6063 cathodes**, **air agitation + cooling.** These let you sanity-check almost anything about this tank.

⚙ **TECHNICAL STUFF — THE “720 RULE” (THICKNESS MATH)**

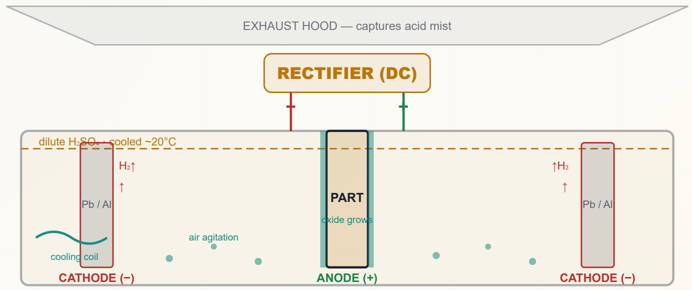
A handy field rule for Type II: at about **12 ASF**, you grow roughly **0.0001” (≈2.5 µm) every ~4–5 minutes**. A common shop version is the **“720 rule”:** *minutes = (720 × desired thickness in mils) ÷ current density in ASF.* So 1.0 mil at 12 ASF ≈ 60 min. Your shop’s exact constant comes from its own data — but the principle is **thickness is set by current density × time.** Verify with an eddy-current gauge (Part Three).

⚙ **TECHNICAL STUFF — WHY TEMPERATURE IS KING**

Warm bath → the acid dissolves the growing oxide too fast → a **soft, powdery, thin, or “burned”** coating with poor wear and poor dyeing. Slightly colder → a **harder, denser** coating (this is exactly how Type III hardcoat is run — colder and harder). For Type II, hold ~20–21 °C tightly. **If cooling fails, the bath quietly ruins every part — watch it.**

⚙ **TECHNICAL STUFF — DISSOLVED ALUMINUM AND CHLORIDE**

A *little* dissolved aluminum helps conductivity and color consistency, but **too much** dulls the coating, slows growth, and causes streaks — the bath eventually needs partial decant/replacement. **Chloride** (from tap water, hand sweat, or drag-in) is the silent killer: even tens of ppm can cause **pitting and burning.** Use DI water and keep contamination out.



Key params: `H2SO4 165-200 g/L · 20-21C (68-72F) · 12-18 V · 12-18 ASF · 5-25 um · air agit + cooling`. Note the **reversed polarity vs. a plating cell:** the part is the ANODE (+).

• STANDARD OPERATING PROCEDURE

1. **Confirm the part is clean, etched, desmutted, rinsed, and racked tight** — straight from the rinse, no long air dwell.
2. **Verify engineering controls FIRST:** cooling running and at temperature (~20–21 °C), air agitation on, exhaust on, splash control in place, full acid PPE on.
3. **Check bath readiness:** sulfuric in range (per lab), temperature in range, dissolved-aluminum and chloride within limits, cathodes present/sound.
4. **Confirm rectifier polarity: PART = ANODE (POSITIVE).** *(This is the opposite of plating — verify it every time.)*
5. **Set current for the part's surface area** to hit the target current density (~12–18 ASF).
6. **Calculate the time** for target thickness (current density × time; the "720 rule") and set the timer/amp-hour counter.
7. **Energize and ramp** per procedure; watch that **voltage rises** to hold current as the insulating oxide grows (that voltage rise is normal).
8. **Anodize for the calculated time**, monitoring temperature, current/voltage, agitation, and cooling the whole run.
9. **At target, stop current, withdraw, drain over the tank,** and move to the post-anodize rinse — the pores are now **open and ready for dye/seal.**
10. **Log everything:** time, current/voltage, temperature, thickness checks, additions.

> ⚠ **HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME**
>
> Because everyone trained on plating expects the part to be the **cathode**, the most dangerous habit on this line is assuming polarity. **The part is the ANODE (+).** Reversed polarity gives no coating (and can damage parts/cathodes). Check it before you energize.

COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Powdery, soft, chalky coating	Bath too warm; CD too high ("burning")	Check cooling/temp & CD; verify agitation
Thin / no coating	Bad contact; wrong polarity; low current; low acid	Re-rack tight; **verify part = anode**; check current/acid
Burned/rough at edges & contact	Local CD too high; poor contact; chloride	Improve racking; check chloride; adjust CD
Pitting	**Chloride contamination**; bath impurities	Lab-check chloride; use DI; treat bath per chemist
Dull, gray, streaky	High-Si/Cu alloy; high dissolved Al; smut not removed	Confirm alloy; lab-check Al; verify desmut
Color won't match part-to-part	Mixed alloys; inconsistent dwell/temp; thickness variation	Rack like-with-like; standardize timing; verify thickness
Voltage spikes / won't hold current	Poor contact; failing cathodes; low conductivity	Check contacts/cathodes; lab-check bath

INTEGRATED SAFETY — THIS STATION DEMANDS TOTAL RESPECT

⚠ **HEADS-UP — SULFURIC ACID (THE HEADLINE)**
Corrosive, splash and **acid mist** hazard. Engineering controls (exhaust, splash guards) first, PPE always (sealed goggles, face shield, gloves, apron, boots). Any skin/eye contact → flush **15 minutes**, get medical attention. **Acid to water** for make-up.

⚠ **HEADS-UP — HIGH-CURRENT DC & FLAMMABLE HYDROGEN**
Shock/arc at contacts and bus bars — **LOTO** for all electrical work, dry hands only. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.

⚠ **HEADS-UP — WATCH THE COOLING THE WHOLE RUN**
A run is 30–60 minutes and the bath self-heats the entire time. If cooling fails or the temperature climbs out of range mid-run, the coating **burns/powders** and the bath degrades. Don't set-and-forget.

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge? *(Anode, +.)*
- Where does the coating come from? *(The part's own Al + O_2 — not the cathodes.)*
- The headline numbers (acid, temp, V, CD, thickness, time)?
- Why must the bath be cold, and what if it warms?
- What does chloride do, and how do you keep it out?
- How is thickness controlled? *(CD × time.)*
- The three safety families? *(Acid, DC, hydrogen.)*

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it's the **anode**).
- Running with failing/insufficient cooling.
- Loose rack contact (no-coat / burned contact).
- Ignoring chloride / dissolved-Al limits.
- Mixing alloys on a rack and expecting matched color.
- Long, variable dwell before anodizing.
- Eating/drinking/face-touching near the acid tank.

SIGN-OFF — STATION 07

"I can independently: verify cooling, agitation, exhaust, and PPE before running; verify bath readiness (acid, temp, dissolved Al, chloride per lab); confirm the part is racked as the ANODE (positive); set the correct current density; control thickness by current × time; recognize burning/pitting/thin-coat defects; and state the acid, electrical, and hydrogen hazards. I understand chemistry adjustments are made only by authorized personnel."

Operator: ____________________ Date: __________ Trainer: ____________________ Date: __________

STATION 08 OF 12

RINSE (POST-ANODIZE)

“GENTLE NOW — THE PORES ARE WIDE OPEN.”

PURPOSE & THE “WHY”

Rinse the sulfuric acid out of the freshly anodized part **without contaminating or prematurely affecting the open pores**, before dye or seal. The fresh anodic oxide is **porous and absorbent** — it will soak up whatever it touches. Leftover acid in the pores interferes with dyeing (off-color, blotchy) and sealing. But the rinse must be **clean (DI)** and not too warm, because warm water can begin to seal the pores early and **block dye uptake.**

★ **REMEMBER — DON’T PRE-SEAL BY ACCIDENT**

After anodizing, **use cool/ambient DI rinse before dyeing.** Hot rinse water here can start closing the pores, and a partially sealed part **won’t dye properly.** Save the heat for the actual seal step.

PARAMETER	RANGE	NOTES
Water	DI strongly preferred	Pores absorb impurities
Temperature	Cool / ambient (before dye)	Avoid premature sealing
Time	1–3 min, gentle agitation	Flush acid from pores
Conductivity	Low (per spec)	Confirms acid removed

⚠ **HEADS-UP**

This rinse carries dragged-out **sulfuric acid** — mildly acidic, slip hazard, goggles/gloves. Don’t let acid-laden rinse go untreated to drain (Part Three / waste).

STEP-BY-STEP

- Confirm DI flow, cool temperature.
- Rinse gently, full time.
- Drain.
- To dye (or, if no color, straight to seal).

TOP MISTAKES

- Using hot water before dye (pre-seals pores).
- Using hard tap water (impurities in pores).
- Short rinse leaving acid behind.

SIGN-OFF — STATION 08

Teach-Back: Why the pores are absorbent now; why cool DI before dye; what hot water would do prematurely.

“I rinsed with cool DI to flush acid from the open pores without pre-sealing, and moved promptly to dye or seal.”

Trainee: ____________ Trainer: ____________ Date: ________

STATION 09 OF 12 · OPTIONAL

DYE (OPTIONAL)

"COLOR GOES INTO THE COATING — THIS IS THE STEP THAT MAKES ANODIZING MAGIC."

PURPOSE & THE "WHY"

Draw **color down into the open pores** of the fresh anodic oxide, coloring the coating from the inside (only if a color is specified — clear parts skip straight to seal). The porous oxide acts like a sponge. Immerse it in a **dye solution** and the dye is absorbed into the pores, becoming part of the coating — far more durable than surface paint. Dye depth depends on **coating thickness** (more pores to fill), **dye concentration, temperature, time, and pH.** After dyeing, the part **must be sealed** to lock the color in.

★ **REMEMBER — DYE THEN SEAL, NEVER THE REVERSE**

Dye only works while the pores are open. **Anodize** → **rinse** → **DYE** → **rinse** → **SEAL.** Once sealed, the part can't be dyed; once dyed, it must be sealed or the color will rub/bleed and fade.

PARAMETER	RANGE / PRACTICE	NOTES
Dye type	**Organic** (acid or basic); or inorganic/electrolytic	Organic = bright/wide gamut; electrolytic = durable bronze/black
Concentration	`Per dye TDS`	Higher/longer = deeper shade
Temperature	`~50–60 °C typical`	Affects uptake rate
pH	`Per dye (often slightly acidic)`	Critical for shade & consistency
Time	`~5–15 min`	Until target shade
Water	`DI`	Tap-water impurities ruin color
Coating thickness	`Adequate for the shade`	Darks need more thickness/pores

⚠ **HEADS-UP — DYE HANDLING**

Some dyes are **skin/eye irritants or sensitizers**, and dye baths may be warm and mildly acidic. Wear gloves and goggles, avoid skin contact, and read each dye's SDS (some carry their own disposal requirements as colorants/heavy-metal content).

💡 **TIP — CONSISTENCY IS EVERYTHING IN COLOR**

Color depends on coating thickness, dwell, dye temperature, pH, and the **interval between anodize and dye.** Keep all of these consistent part-to-part or you'll get shade variation across a lot. **Dye promptly** after the post-anodize rinse — pores age and dye less well if the part sits.

⚙ **TECHNICAL STUFF — LIGHT-FASTNESS AND ELECTROLYTIC COLOR**

Organic dyes give a huge color range but some **fade in UV/outdoors** (check ratings for exterior work). For durable architectural bronzes and blacks, shops use **two-step electrolytic coloring** (depositing tin/nickel/cobalt metal salts deep in the pores), which is extremely light-fast — that's its own manual in this series.

STEP-BY-STEP

- Confirm dye type, concentration, temp, pH (DI-based).
- Confirm part came from a cool DI rinse (pores open).
- Immerse fully, no trapped air; agitate gently to target shade.
- Drain, then **rinse** off surface dye.
- Move promptly to **seal.**

TOP MISTAKES

- **Sealing before dyeing** — fatal order error.
- Inconsistent dwell/temp/pH → shade variation.
- Letting parts sit between anodize and dye.
- Hard/tap water in the dye line.
- Not rinsing surface dye → bleeding after seal.
- Expecting a bright shade on thin coat / high-Si/Cu alloy.

SIGN-OFF — STATION 09

Teach-Back: Where the color goes (into open pores) and why that's durable; the correct order (anodize → rinse → dye → rinse → seal); variables that control shade; why DI/consistency matter; the dye-handling hazards.

"I dyed before sealing, into open pores, with consistent parameters in DI water, rinsed off surface dye, and moved promptly to seal."

Trainee: ____________________ Trainer: ____________________ Color/shade: ____________

STATION 10 OF 12

SEAL (CLOSE THE PORES)

"THE FINAL LOCK — CLOSE THE PORES TO KEEP THE COLOR IN AND THE CORROSION OUT."

PURPOSE & THE "WHY"

Close the open pores of the anodic oxide so the coating reaches its full corrosion resistance and any dye is locked in permanently. An unsealed anodize is porous — it can absorb stains, lose dye, and corrode through the pores. **Sealing** swells/plugs the pore mouths shut. The classic method is **hot DI water near boiling**, which converts the pore-wall oxide to a swollen hydrated oxide (boehmite) that closes the pores. There are also **mid-temperature nickel-acetate** seals and **cold/room-temperature nickel-fluoride** seals — each trading energy, speed, and performance.

★ **REMEMBER — SEALING IS THE POINT OF NO RETURN**

After sealing you **cannot dye** (pores are shut). Seal is the *last* chemical step. A poorly sealed part will fade, stain, and corrode; an over-sealed/contaminated part can show **"sealing bloom"** (a chalky/smudgy film). Sealing is where a beautiful job is finished — or quietly ruined.

THE THREE COMMON SEALS

SEAL TYPE	TEMPERATURE	TRADEOFFS
Hot DI water seal	~96–100 °C	Simple, no metals, excellent quality; high energy, slow (~2–3 min/µm), prone to **bloom** if water is off
Nickel-acetate (mid-temp)	~80–90 °C	Lower energy; excellent corrosion & dye retention; widely used; contains **nickel**
Cold nickel-fluoride	~25–30 °C	Lowest energy/fastest; often a warm-water "aging" rinse after; contains **nickel + fluoride**; needs tight control

⚙ **TECHNICAL STUFF — WHY HOT WATER SEALS**

Near-boiling water hydrates the oxide at the pore walls to **boehmite (AlOOH)**, which occupies more volume than the original oxide and **swells the pore mouths shut.** Nickel seals add nickel hydroxide/salts that precipitate in the pore mouths to help plug them, which is why nickel-acetate works at lower temperature and cold nickel-fluoride can work near room temperature. All three accomplish the same goal — **closed pores** — by different routes.

⚙ **TECHNICAL STUFF — "SEALING BLOOM" / SMUT**

A powdery white/gray film after sealing ("bloom") usually comes from **hard water, drag-in, wrong pH, or over-concentrated seal** depositing on the surface. It's often removed with a light acid or proprietary post-dip — but the real fix is **clean DI water and correct seal chemistry/pH/temperature.** Bloom on a dyed part can dull the color.

⚠ HEADS-UP — NEAR-BOILING TANK = SCALD HAZARD

A hot-water or hot nickel seal runs **~80–100 °C.** Splashes and steam cause **serious scald burns.** Use a face shield, heat-resistant gloves, long sleeves, and mind the steam cloud. Lower parts gently — a fast plunge throws hot water.

⚠ HEADS-UP — NICKEL SEALS ARE A HEALTH & WASTE CONCERN

Nickel is a skin sensitizer and the **fluoride** in cold seals is especially hazardous (deep, delayed burns; systemic toxicity). Gloves and goggles, read the SDS, know the fluoride first-aid (**calcium gluconate** per your plan), and route nickel/fluoride seal waste to proper treatment — **never to drain.**

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Water	`DI (hot-water seal)`	Hard water → bloom
Temperature	`Hot ~96–100 · Ni-acetate ~80–90 · Ni-F ~25–30 °C`	Per chosen seal
pH	`Per seal product (often ~5.5–6.5)`	Wrong pH = poor seal/bloom
Time	`~2–3 min per µm (hot water)`	Per seal type & thickness
Result	Closed pores; passes seal-quality test	Dye-stain or admittance test (Part Three)

STEP-BY-STEP

1. Confirm seal type, temperature, pH, and DI water are correct/in range. 2. Confirm the part came from a **clean rinse** after dye (no surface dye/contamination). 3. Immerse gently (mind the scald), full time for the coating thickness. 4. Withdraw, rinse if specified, and inspect for **bloom.** 5. Move to final rinse/dry.

TEACH-BACK — CAN YOU ANSWER?

- What sealing does (closes pores) and why (lock dye + corrosion).
- Why you can't dye after sealing.
- The three seal types and one tradeoff of each.
- The scald hazard of hot seals; the nickel/fluoride hazard of nickel seals.
- What sealing bloom is and its usual cause.

TOP MISTAKES

- **Trying to dye after sealing** (impossible).
- Wrong temp/pH → incompletely sealed (fades/corrodes).
- Hard/tap water → sealing bloom.
- Plunging into a near-boiling tank (scald + shock).
- Routing nickel/fluoride seal waste to drain.
- Skipping the seal-quality check.

SIGN-OFF — STATION 10

"I sealed at the correct type/temperature/pH in DI water after dyeing, checked for bloom, and handled the scald and (if applicable) nickel/fluoride hazards correctly."

Trainee: ____________ Trainer: ____________ Seal type/temp: ________

STATION 11 OF 12

RINSE / DRY

"FINISH CLEAN, DRY GENTLY, DON'T UNDO YOUR WORK."

PURPOSE & THE "WHY"

Give the sealed part a final clean rinse and a **gentle dry** without water spotting or thermal damage. A clean final rinse removes any seal-tank drag-out so the part dries spot-free. Drying should be **gentle** — a sealed, dyed part is finished, and aggressive heat or hard-water spotting at the very end can blemish an otherwise perfect job.

PARAMETER	RANGE	NOTES
Final rinse water	`DI`	Spot-free drying
Rinse temperature	`Warm`	A warm final rinse aids drying
Dry method	`Air / warm forced air; gentle oven`	Avoid excessive heat
Handling	`By rack / clean gloves`	No bare-hand prints on finished parts

⚠ **HEADS-UP**

Parts coming off a hot seal are **hot** — burn risk. Let them cool or use gloves. Dry ovens are a heat hazard; mind hot racks.

💡 **TIP**

A **DI final rinse** is the cheapest insurance against water spots on a finished, dyed part. Tap-water spots on a glossy clear or black anodize are obvious and are pure rework.

STEP-BY-STEP

- Final DI rinse.
- Drain.
- Gentle dry.
- Handle by rack/gloves to inspection.

TOP MISTAKES

- Tap-water final rinse (spots).
- Over-aggressive drying.
- Bare-hand handling of finished parts.

SIGN-OFF — STATION 11

Teach-Back: Why a clean final rinse + gentle dry protects the finished job.

"I gave a clean DI final rinse, dried gently without spotting/overheating, and handled finished parts cleanly."

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 12 OF 12

FINAL INSPECTION

"PROVE IT'S RIGHT — THICKNESS, SEAL, COLOR, AND FINISH."

PURPOSE & THE "WHY"

Verify the finished part meets spec: **coating thickness, seal quality, color match, and overall appearance.** Anodize quality isn't visible by eye alone. **Thickness** (eddy-current gauge) confirms you grew enough oxide; **seal quality** (dye-stain or admittance test) confirms the pores actually closed; **color** confirms the dye took correctly; and a visual check catches burns, bloom, contact marks in wrong places, and handling damage.

CHECK	METHOD	PASS CRITERIA
Coating thickness	**Eddy-current gauge** on significant surfaces	Within spec (e.g., 5–25 µm per print)
Seal quality	**Dye-stain** (ASTM B136) and/or **admittance** (ASTM B457)	Low dye pickup / low admittance = well sealed
Color	Visual / spectro vs. standard	Matches approved sample
Appearance	Visual under good light	No burns, bloom, pits, streaks, bad contact marks
Dimensions	Gauging where critical	Within tolerance **after** dimensional growth

★ **REMEMBER — THE TWO TESTS THAT MATTER MOST**

Eddy-current = "did we grow enough oxide?" and the **seal test (dye-stain/admittance) = "did we close the pores?"** A part can be the right color and the right thickness and *still fail* if it isn't sealed — which is why the seal test exists.

💡 **TIP — THE DYE-STAIN SEAL TEST IN PLAIN TERMS**

Drop a spot of test dye on a well-sealed area; a **good seal barely picks up color** (wipes clean), a **poor seal stains** (pores still open and grabbing dye). It's the quick visual cousin of the more precise **admittance** test, which measures how electrically "open" the coating still is.

STEP-BY-STEP

- Eddy-current thickness on significant surfaces.
- Seal test (dye-stain/admittance).
- Color vs. standard.
- Visual for defects.
- Gauge critical dimensions.
- Accept / reject / route to rework (strip & re-anodize).

TOP MISTAKES

- Measuring thickness only on easy (not significant) surfaces.
- Skipping the seal test.
- Judging color under bad light.
- Missing dimensional growth on critical fits.

SIGN-OFF — STATION 12

Teach-Back: Which check answers "enough oxide?" vs. "pores closed?"; what a dye-stain test result means.

"I verified thickness on significant surfaces, ran the seal-quality test, checked color against the standard, inspected for defects, and confirmed critical dimensions accounting for growth."

Trainee: ____________________ Trainer: ____________________ Thickness: ________ µm · Seal: P/F

PART THREE — ACROSS THE LINE
THE CROSS-CUTTING TOPICS

ACROSS THE LINE

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

• COATING THICKNESS & DIMENSIONAL GROWTH (DEEP DIVE)

Thickness is set by current density × time. At a fixed current density (~12–18 ASF), the longer you anodize, the thicker the oxide — within limits (eventually the acid's dissolution rate caps practical thickness for Type II). The shop rule (the "720 rule") ties minutes, mils, and ASF together: *minutes ≈ (720 × mils) ÷ ASF.* Always **verify with an eddy-current gauge** on the **significant surface**, not a convenient corner.

Dimensional growth is the operator-critical consequence: oxide is bulkier than the metal it came from, so the coating grows **~50% in, ~50% out.** Each surface moves out ~half the coating thickness; a **diameter changes by ~the full thickness** (pins grow, bores shrink). Tight-tolerance, threaded, and press-fit features must be **machined to allow for growth** or **masked off.**

> ★ **REMEMBER**
> Two numbers rule this section: **thickness = CD × time** (verify by eddy-current) and **growth ≈ half-in/half-out** (plan tolerances and masking around it).

• RACKING & ELECTRICAL CONTACT (DEEP DIVE)

The part is the anode, so **the rack is the wire that grows your coating.** Good racking means: **firm, current-carrying contact**; **contact placed where the bare mark won't matter**; **orientation for gas escape and drainage** (no trapped air = no un-anodized voids); **no parts touching** (shadowing/marks); and **maintained racks** (titanium preferred; aluminum racks must be stripped/re-pointed as they anodize over). Poor contact is the single most common cause of thin coatings, burned spots, and dropped parts. *(Cross-reference Station 01.)*

• DYEING & SEALING (DEEP DIVE)

Dye lives *in* the pores; its shade depends on **coating thickness, dye concentration/temperature/pH/time, the anodize-to-dye interval, alloy, and DI water quality.** Organic dyes give a wide gamut but vary in **light-fastness** (check ratings for outdoor parts); **two-step electrolytic** color gives extremely durable architectural bronzes/blacks. **Seal** closes the pores. Three routes: **hot DI water (~96–100 °C)** — simple, metal-free, high energy, bloom-prone if water is off; **nickel-acetate mid-temp (~80–90 °C)** — efficient, excellent corrosion/dye retention, contains nickel; **cold nickel-fluoride (~25–30 °C)** — fastest/lowest energy, but nickel + fluoride hazard and tight control needed. The universal enemies are **hard water, wrong pH, and contamination**, which cause **sealing bloom.** Confirm the seal worked with a **dye-stain (ASTM B136)** or **admittance (ASTM B457)** test.

> ★ **REMEMBER**
> **Dye = color into open pores. Seal = pores closed forever.** Order is anodize → dye → seal, never reversed. Test the seal — color and thickness alone don't prove it.

• ALUMINUM ALLOY EFFECTS (DEEP DIVE)

The alloy decides clarity and color. **Wrought 5xxx/6xxx (esp. 6063)** anodize bright and dye true. **Copper (2xxx)** dulls and yellows. **Silicon (cast alloys)** turns the coating **gray/charcoal** and mottled. Rack **like alloy with like alloy** to keep color consistent, and anodize a **sample of the actual alloy/lot** before promising a tight color match. *(Cross-reference Chapter 6.)*

• RINSING & WATER/DI MANAGEMENT

Rinses are firewalls between opposite chemistries (caustic etch vs. acid desmut/anodize) and protectors of the dye/seal. **Counter-current cascade** rinsing saves water; **conductivity** is the cleanliness gauge (lower = cleaner). Use **DI water** for the post-anodize rinse, dye, seal, and final rinse — tap-water impurities (especially **chloride** and hardness) cause pitting, off-color dye, and sealing bloom.

• WASTE & ENVIRONMENTAL

An anodize line generates several regulated streams — **never to a floor drain untreated:**

- **Spent sulfuric acid & acid rinses** — neutralize/treat per permit; manage **dissolved aluminum** (precipitated as aluminum hydroxide sludge).
- **Spent caustic etch** — high pH; contains dissolved aluminum; neutralize/treat.
- **Desmut waste** — acidic; **fluoride-bearing** desmuts need fluoride treatment (precipitate as calcium fluoride).
- **Dye waste** — colorants; some contain regulated metals; treat/dispose per SDS and permit.
- **Nickel-seal waste** (nickel-acetate / nickel-fluoride) — **nickel is regulated**; fluoride needs treatment; segregate and treat — never to sewer.
- **Air** — acid mist (sulfuric) and steam need exhaust/scrubbing per permit.

> ⚠ **HEADS-UP**
> Acidic and caustic streams must be **neutralized under control** (never casually mixed), and nickel/fluoride streams need **dedicated treatment.** Know your shop's segregation scheme; the wrong drum or drain can be a reportable violation.

• QUALITY CHECKS & TEST METHODS

CHECK	METHOD (TYPICAL STD)	ANSWERS
Coating thickness	Eddy-current gauge (ASTM B244); micro-section if needed	“Enough oxide grown?”
Seal quality	Dye-stain (ASTM B136); admittance (ASTM B457); acid-dissolution weight loss	“Pores closed?”
Color	Visual / spectrophotometer vs. standard	“Right shade & match?”
Abrasion / wear	Taber abrasion / per spec	“Hard/durable enough?”
Corrosion	Salt spray (ASTM B117) per spec	“Will it survive service?”
Appearance/defects	Visual under controlled light	Burns, bloom, pits, streaks, marks

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Burning (powdery, dark, rough)	Bath too warm; CD too high; poor agitation	Restore cooling/agitation; lower CD; verify temp
Powdery / soft / chalky	Overheated bath; over-aggressive conditions	Cool bath to ~20–21 °C; check CD & agitation
Thin or no coating	Poor contact; **wrong polarity (part not anode)**; low current; low acid	Re-rack tight; **verify part = anode**; check current/acid
Pitting	**Chloride contamination**; impurities	Lab-check chloride; switch to DI; treat bath
Color variation (part-to-part)	Mixed alloys; inconsistent thickness/dwell/dye temp/pH	Rack like-with-like; standardize timing & dye conditions
Dull / gray / muddy color	High-Si or high-Cu alloy; high dissolved Al; smut left on	Confirm alloy; lab-check Al; verify desmut
Color too light / won't go dark	Coating too thin (too few pores); weak/old dye; short dye time	Increase thickness; refresh dye; lengthen dye dwell
Poor seal (fades, stains, fails test)	Wrong seal temp/pH; contaminated seal; partially pre-sealed before dye	Correct temp/pH/DI; re-seal; don't pre-seal before dye
Sealing bloom (chalky film)	Hard water; wrong pH; over-concentrated seal	DI water; correct pH/concentration; light post-dip
Streaks / blotchy finish	Smut not removed; uneven cleaning/etch; drag-in	Verify clean/etch/desmut; consistent dwell
Contact burns / no-coat at contact	Loose/insufficient rack contact; current crowding	Tighten/add contacts; re-point/replace rack
Color bleeding after seal	Surface dye not rinsed before seal	Rinse off surface dye before sealing
Gray cast on cast parts	High-silicon alloy (inherent)	Set expectations; consider different alloy/finish

💡 **TIP**

Most anodize defects trace to one of four roots: **temperature/cooling** (burning), **contamination** (chloride/hard water — pitting & bloom), **alloy** (gray/dull), or **order/handling** (dye-seal order, contact, dwell). Name the root, not just the symptom.

★ **REMEMBER**

Two of these are **safety-relevant** as well as quality: **wrong polarity** (verify the part is the anode) and any move that increases acid attack or heat. When in doubt about a bath change, ask your supervisor before you act.

EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

Admittance test: an electrical test (ASTM B457) of how "open" the coating still is — used to confirm sealing (lower admittance = better seal).
Anode: the **positive** electrode. In anodizing it's **your part** — where the oxide grows.
Anodizing: growing an integral oxide layer out of a metal's surface by making the part the anode in an electrolyte.
Barrier layer: the thin, dense oxide at the bottom of the coating, against the metal (under the porous layer).
Boehmite: hydrated aluminum oxide (AlOOH) formed during hot-water sealing; its swelling closes the pores.
Cathode: the **negative** electrode (lead or aluminum here) — carries current and bubbles off hydrogen; does **not** feed the coating.
Chloride contamination: dissolved chloride in the bath that causes **pitting/burning**; kept very low (often ≤ ~50 ppm).
Current density (ASF / A/dm²): amps per unit area. Type II ≈ **12–18 ASF (~1.2–2.0 A/dm²)**.
Desmut / deox: an acid dip that removes the dark **smut** left by caustic etching.
Dimensional growth: the coating grows **~half in, half out**; each surface moves out ~half the thickness.
Drag-out / drag-in: solution clinging to a part leaving a tank (out) / contaminating the next tank (in).
Dye: organic or inorganic colorant absorbed **into the open pores** of fresh anodic oxide.
Eddy-current gauge: non-destructive instrument for measuring coating thickness on aluminum.
Etch: a hot **caustic** dip that removes the natural oxide and a little aluminum, giving the satin finish.
Integral coating: a coating that *is* part of the metal (grown from it), not added on top — can't peel to bare metal.
MIL-PRF-8625 (MIL-A-8625): the spec defining anodize **Type I (chromic), II (sulfuric), III (hardcoat)** and classes.
Porous layer: the thick upper oxide full of vertical **pores** — what makes dyeing possible.
Sealing: the final step that **closes the pores** (hot water, nickel-acetate, or cold nickel-fluoride) to lock in color and corrosion resistance.
Sealing bloom (seal smut): a chalky film after sealing, usually from hard water / wrong pH / contamination.
Significant surface: the surface that must meet thickness/finish spec; where you measure.
Smut: dark residue of alloying elements (Cu, Si, Fe) left on the surface after caustic etching.
Type II: the standard **sulfuric** anodize — medium thickness, dyeable, decorative + protective.
Valve metals: metals that anodize to form protective oxides (aluminum, titanium, magnesium, tantalum, niobium).
Water-break test: a free cleanliness check — clean metal sheets water; dirty metal beads it.

★ **REMEMBER**
Of all these, five matter most: **anode** (your part), **porous oxide** (why dyeing works), **dye-then-seal** (the unique heart of anodizing), **dimensional growth** (plan your tolerances), and **chloride/temperature control** (burning & pitting). Master those and you understand Type II.

☀ **FUN FACT**
The word **"anode"** comes from the Greek *ánodos*, "a way up" (Faraday's term for the electrode where current enters the cell). On this line, that "way up" is literally where your coating grows — a fitting name for the star of the show.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	**Safety / PPE / SDS / hygiene — acid, caustic, hot seal, hydrogen, DC (all stations)**	____	________	________	______
2	Station 01 — Inspection & Racking (contact)	____	________	________	______
3	Station 02 — Clean / Degrease + water-break	____	________	________	______
4	Station 03 — Etch (caustic)	____	________	________	______
5	Station 04 — Rinse + conductivity	____	________	________	______
6	Station 05 — Desmut / Deox	____	________	________	______
7	Station 06 — Rinse	____	________	________	______
8	Station 07 — Anodize (part = anode) operation	____	________	________	______
9	Thickness control (CD × time) & dimensional growth	____	________	________	______
10	Station 08 — Post-anodize rinse (open pores)	____	________	________	______
11	Station 09 — Dye	____	________	________	______
12	Station 10 — Seal (close pores)	____	________	________	______
13	Station 11 — Rinse / Dry	____	________	________	______
14	Station 12 — Final Inspection (thickness/seal/color)	____	________	________	______
15	LOTO on rectifier / electrical safety	____	________	________	______
16	Emergency response — eyewash, shower, spill, fluoride first-aid	____	________	________	______
17	Waste segregation & environmental controls	____	________	________	______
18	Troubleshooting — symptom recognition	____	________	________	______

Operator name: ______________________________ **Hire/start date:** ____________________

Supervisor name: ______________________________ **Review cycle:** ☐ Annual ☐ Other: ______________

⚠ **HEADS-UP**

Safety rows (1, 15, 16, 17) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS/hygiene, LOTO, emergency response (including **fluoride first-aid** where used), and waste-control awareness are signed. Safety competency precedes production competency.

PART FOUR — CERTIFICATION
20 QUESTIONS · PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**
Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q3, Q9, Q13, Q17, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ____________ **Score:** ______ / 20 **Safety-gate Qs: 3, 9, 13, 17, 19**

Q1. Anodizing differs from electroplating mainly because in anodizing the part is the:
A. Cathode, and a different metal is deposited on it B. Anode, and an oxide is grown out of the part itself C. Insulator, and nothing happens D. Cathode, and it dissolves

Q2. The Type II anodize electrolyte is:
A. Chromic acid B. Sodium hydroxide C. Dilute sulfuric acid (~165–200 g/L) D. Nickel sulfate

Q3. **[SAFETY GATE]** The single most important reason the anodize bath must be **cooled** to ~20–21 °C is:
A. To save electricity B. Temperature doesn't matter C. To make it freeze D. Because the reaction is exothermic and a warm bath gives a soft/powdery/“burned” coating (and the hot acid is a hazard)

Q4. Typical Type II coating thickness is about:
A. 0.5–1 mm B. 5–25 µm (0.0002–0.001″) C. 1–2 inches D. There is no coating

Q5. The anodic oxide is **porous.** Why does that matter?
A. The pores let us dye the coating, then we must seal them B. It makes the part weaker only C. It has no effect D. It means the part is dirty

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is and **where the coating comes from.**

Q7. “Dimensional growth” in anodizing means the coating grows roughly:
A. All outward B. All inward C. Half into the surface and half out (each surface moves out ~half the thickness) D. It shrinks the part

Q8. The correct process order for a colored part is:
A. Seal → dye → anodize B. Dye → seal → anodize C. Anodize → dye → seal D. Dye → anodize → seal

Q9. (Short answer) **[SAFETY GATE]** Name the **two corrosive hazards** on the line (one high-pH, one low-pH), and state the universal rule for adding acid during bath make-up.

Q10. Typical Type II voltage and current density are about:
A. 12–18 V; 12–18 ASF B. 200 V; 500 ASF C. 1 V; 0.1 ASF D. 1000 V; 1 ASF

Q11. A part comes out **gray and mottled** even though the line is running fine. The most likely cause is:
A. The bath is too clean B. The rectifier is off C. Too much dye D. A high-silicon (cast) or high-copper (2xxx) alloy

Q12. (Short answer) What is **smut**, where does it come from, and which station removes it?

Q13. [SAFETY GATE] The seal tank in a hot-water or nickel-acetate seal runs at about **80–100 °C.** The main hazard is:
A. Scald/steam burns B. Electrical shock C. Radiation D. None

Q14. Why can't you **dye** a part after it has been **sealed**?
A. The dye is too expensive B. Sealing closes the pores, so there's nowhere for the dye to go C. You can — order doesn't matter D. The part is too cold

Q15. Coating **thickness** in Type II is controlled mainly by:
A. Dye color B. The alloy only C. Seal temperature D. Current density × time

Q16. (Short answer) Name the **three** common seal methods and give **one tradeoff** of each.

Q17. (Short answer) [SAFETY GATE] List **three** pieces of PPE required at the sulfuric anodize/desmut tanks, and name the special first-aid concern if your desmut or cold seal contains **fluoride.**

Q18. Chloride contamination in the anodize bath causes:
A. Brighter color B. Pitting / burning of the coating C. Faster sealing D. Nothing

Q19. (Short answer) [SAFETY GATE] Name the **flammable gas** produced at the cathode during anodizing and **two** controls that keep it from becoming a fire hazard.

Q20. (Short answer) Name **two** ways sulfuric anodizing is fundamentally different from electroplating zinc or chrome.

⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**

Questions 3, 9, 13, 17, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

⚠ **HEADS-UP**

Keep this page out of the trainee's copy. Questions **3, 9, 13, 17, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Answer distribution (MC): A=3 · B=4 · C=3 · D=3

Q1. B — The part is the **anode**, and oxide is grown out of the part itself.

Q2. C — Dilute sulfuric acid (~165–200 g/L, ~15–20% w/v).

Q3. D — The reaction is **exothermic**; a warm bath dissolves the oxide too fast → soft/powdery/"burned" coating, and hot strong acid is a hazard. (Cooling + agitation hold ~20–21 °C.)

Q4. B — ~5–25 µm (0.0002–0.001″).

Q5. A — The pores allow **dyeing**, and then they **must be sealed** to lock in color/corrosion resistance.

Q6. In plating the part is the **cathode** and a **different metal is deposited onto it** from the bath. In anodizing the part is the **anode**, and the coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath combines with the part's aluminum) — nothing is deposited from elsewhere.

Q7. C — Half in, half out; each surface grows out ~half the coating thickness (a diameter changes by ~the full thickness).

Q8. C — Anodize → dye → seal.

Q9. The **caustic etch** (high pH / sodium hydroxide) and the **sulfuric anodize/desmut** (low pH / acid) — both cause severe burns. Universal rule: **always add ACID to WATER, never water to acid.**

Q10. A — ~12–18 V; ~12–18 ASF (~1.2–2.0 A/dm²).

Q11. D — A **high-silicon cast alloy** (gray/mottled) or a **high-copper 2xxx alloy** (dull/yellow). The alloy decides the look.

Q12. Smut is the dark film of alloying elements (copper, silicon, iron) **left behind by the caustic etch** when the aluminum dissolves. It's removed at the **desmut/deox** station (acid dip) before anodizing.

Q13. A — Scald/steam burns (the seal runs ~80–100 °C). Use a face shield, heat gloves, mind the steam.

Q14. B — Sealing **closes the pores**, so dye has nowhere to go. Dye must come *before* seal.

Q15. D — Current density × time (verify with an eddy-current gauge).

Q16. Hot DI water (~96–100 °C) — simple/metal-free but high energy and bloom-prone; **nickel-acetate (~80–90 °C)** — efficient/excellent retention but contains nickel; **cold nickel-fluoride (~25–30 °C)** — fast/low-energy but nickel + fluoride hazard and tight control.

Q17. PPE (any three): sealed chemical splash goggles, face shield, chemical-resistant gloves, apron/suit, chemical-resistant boots. **Fluoride concern:** fluoride causes **deep, delayed, systemically toxic burns** — requires specific first-aid (e.g., **calcium gluconate** per SDS/medical plan) and immediate medical attention; ordinary flushing alone may not be enough.

Q18. B — Pitting/burning of the coating. Keep chloride very low (use DI water).

Q19. Hydrogen gas (at the cathode). Controls (any two): **ventilation/exhaust running**, **no open flames/sparks/grinding nearby**, no smoking, keep ignition sources away.

Q20. Any two of: the part is the **anode** (not cathode); the coating is **grown from the part** (not deposited); the coating is **integral aluminum oxide** (not a different metal); it's **porous and dyeable then sealed**; it causes **dimensional growth (half-in/half-out)**; only works on **aluminum/valve metals**; **cathodes don't feed the coating**.

★ **REMEMBER**

A score of 16/20 passes — but every safety question (3, 9, 13, 17, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people.

PRINT, FILL IN, SIGN

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CERTIFICATE OF COMPLETION

SULFURIC ACID ANODIZING (TYPE II) TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Sulfuric Acid Anodizing (Type II)** training course — covering the full anodize workflow (Inspection & Racking, Clean, Etch, Rinse, Desmut/Deox, Anodize, Dye, Seal, Rinse/Dry, and Final Inspection), the fundamental concept that **the part is the anode and the oxide is grown out of the part itself**, the porous-oxide / dye-then-seal principle, dimensional growth, coating-thickness control by current density and time, electrical-contact and racking practice, aluminum-alloy effects, sealing chemistry and seal-quality testing, and — above all — the **safety practices** required of every operator (sulfuric acid and caustic-etch burn control, hot-seal scald control, hydrogen and DC-electrical controls, fluoride first-aid where applicable, spill response, and waste segregation) — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: __________________ **Re-certification due:** __________________

____

TRAINEE SIGNATURE

____

CERTIFYING TRAINER

____

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/aerospace specifications (e.g., MIL-PRF-8625 Type II), and applicable regulatory requirements before production use. Sulfuric acid and caustic etch cause severe burns; seal tanks run near boiling — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.